# TECHNICAL DATA SHEET
## ProLine ABS Filament

### Product Overview

ProLine ABS is an FFF/FDM 3D-printing filament manufactured from ABS-based raw material. The material combines high toughness with strong impact resistance and offers moderate heat resistance, making it a reliable choice for a wide range of FFF 3D printers and functional printed parts.

### Key Features

- Easy processing on common FFF/FDM 3D printers
- High toughness for durable, resilient parts
- Strong impact resistance for functional applications
- Practical heat resistance for everyday engineering use

### Material Properties

| Physical Properties | | | |
|---|---|---|---|
| **Property** | **Test Method** | **Unit** | **Value** |
| Density | ISO 1183 | g/cm³ | 1.04–1.06 |
| MFR (250°C / 2.16 kg) | ISO 1133 | g/10 min | 2-4 |
| Moisture Absorption (23°C / 24 h) | ISO 62 | % | 1% |
| **Mechanical Properties** | | | |
| Tensile strength (X-Y) | ISO 527 | MPa | 35-40 |
| Elongation at break (X-Y) | ISO 527 | % | 12-17 |
| Flexural modulus (X-Y) | ISO 527 | MPa | 1500-1650 |
| Flexural strength (X-Y) | ISO 178 | MPa | 65-70 |
| Impact strength (X-Y) | ISO 180 | kJ/m² | 7-10.5 |
| **Thermodynamic Properties** | | | |
| HDT @ 0.455 MPa (66 psi) | ISO 75 | °C | 88 |

## Recommended Printing Parameters

| Parameter | Setting |
|---|---|
| Print Temperature | 250-270°C |
| Bed Temperature | 90–110°C |
| Bed Materials | Tempered glass, BuildTak, Carbon fiber board |
| Nozzle Diameter | ø 0.4 / 0.6 mm |
| Model Cooling Fan | 0–50% |
| Printing Environmental Temperature | Room temperature to ~40°C |
| Retraction Distance | 1~3mm |
| Supporting Materials | Itself, HIPS |

## Drying & Handling Notes

- Pre-dry: 80°C, minimum 5 h (hot-air oven) to reduce moisture-related printing issues and improve surface/part quality

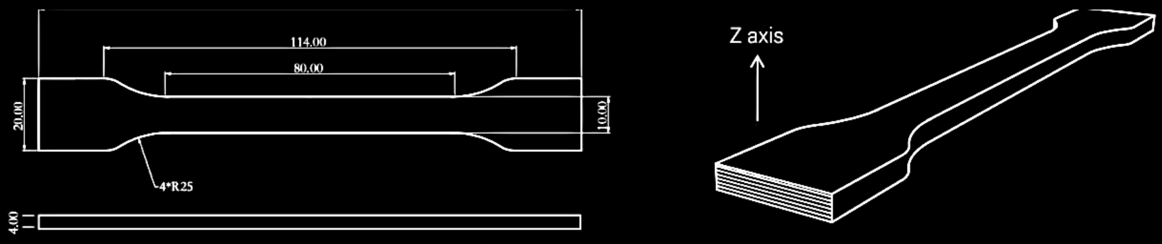
## Disclaimer

Product information and test results in this document were obtained under controlled laboratory conditions. Real-world performance may vary depending on processing, environment, and application. The customer is responsible for determining product suitability for the intended use and for ensuring compliance with all applicable laws and regulations. The Company accepts no liability for the use of this information and provides no warranties, including any implied warranties of merchantability or fitness for a particular purpose, to the extent permitted by law.

## Test Specimens

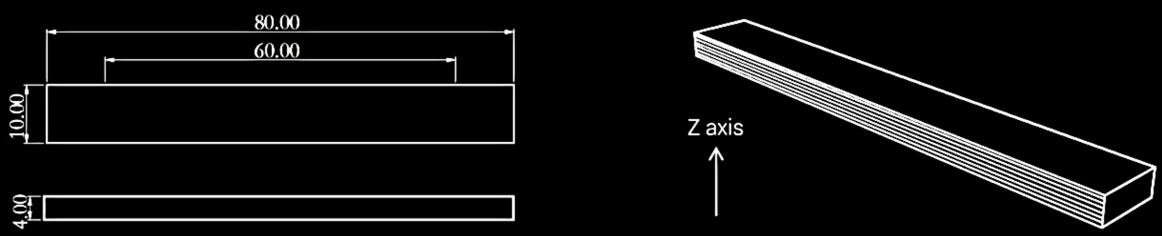
### Test sample dimensions and printing direction

**Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)**

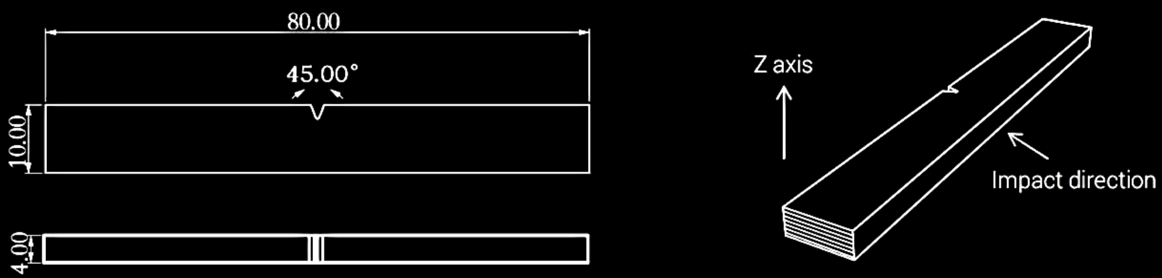



Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

**Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)**




Flexural testing specimen; ASTMD790 (ISO 178, GB/T 9341)




Impact testing specimen; ASTM D256 (ISO 179, GB/T 1043)